

/Trend-Continuation-Agent

Claude Code Implementation Specification

v1.0 · Pepperstone Spread Betting · cTrader MCP HTTP

Field	Detail
Purpose	Multi-market trend continuation scanner with scored confluence, trade card output, and automated order execution via cTrader MCP
Platform	Claude Code (agentic mode) + cTrader MCP via HTTP (primary)
Account	Pepperstone Spread Betting — expandable to CFD with config flag
Instrument scope	33 configured instruments + full available SB symbol list scan
Timeframes	4H (trend / gate analysis) · 1H (entry timing / scoring)
Architecture	Orchestrator + 4 Sub-Agents + Execution Agent
Output	Top-3 Trade Cards with entry, SL, TP1/2/3, confluence detail, Claude commentary
Trade entry	Risk-based position sizing — Pravinder states risk (e.g. '£450 risk') — agent calculates stake and executes

Contents

§	Section
1	Architecture Overview — Orchestrator + Sub-Agent Map
2	Sub-Agent 1 — Data Retrieval Agent
3	Sub-Agent 2 — Gate Layer Agent
4	Sub-Agent 3 — Scoring & Ranking Agent
5	Sub-Agent 4 — Trade Card & Commentary Agent
6	Execution Agent — Order Placement with Risk Sizing
7	Gate Logic — Full Specification
8	Scoring Logic — Full Specification
9	Trade Card — Format & Output Specification
10	Rescan Logic — UK Time / DST Awareness
11	cTrader MCP HTTP — Connection & Instrument Discovery

§1 — Architecture Overview

The agent is structured as an Orchestrator coordinating four specialised Sub-Agents and one Execution Agent. This separation ensures data retrieval, gate analysis, scoring, and output generation each operate independently, allowing Claude Code to parallelise where possible and isolate failures cleanly. The Orchestrator manages state, passes typed payloads between agents, and drives the final output. The Execution Agent is only invoked on explicit user instruction — it never fires automatically.

Layer	Agent	Responsibility	Input	Output
Orchestration	Orchestrator	Controls full scan loop, routes payloads, drives output	User trigger + config	Final trade cards + prompt to execute
Data	Sub-Agent 1 Data Retrieval	Fetches 4H + 1H OHLCV for all instruments via MCP HTTP	Instrument list	Structured OHLCV dict per symbol
Gate	Sub-Agent 2 Gate Layer	Applies 4 binary gates — pass/fail per instrument	OHLCV dict	Passed list with gate metadata
Score	Sub-Agent 3 Scoring & Ranking	Scores 6 signals, assigns tier A/B/C, ranks output	Passed instruments	Ranked scored list with direction
Output	Sub-Agent 4 Trade Card	Builds top-3 trade cards with SL/TP calc and commentary	Ranked list	Formatted trade cards + rescan guidance
Execution	Execution Agent	Places order via MCP with risk-based position sizing	User risk instruction	Confirmed order + MCP response

Agent Communication Pattern

Each sub-agent receives a typed payload from the Orchestrator and returns a typed response. Agents do not call each other directly. The Orchestrator is the single source of truth for scan state. If any sub-agent fails (MCP timeout, empty data), the Orchestrator logs the failure, marks the instrument as skipped, and continues — it does not abort the entire scan.

Step	Action
1	Orchestrator fetches available instrument list from MCP (getSymbols or equivalent)
2	Orchestrator batches instruments → Sub-Agent 1 fetches OHLCV (4H 200 bars, 1H 100 bars)
3	Sub-Agent 1 returns structured data dict → Orchestrator passes to Sub-Agent 2
4	Sub-Agent 2 evaluates 4 gates per instrument → returns {symbol, passed:bool, gates:{}, direction}
5	Orchestrator filters passed instruments → sends to Sub-Agent 3
6	Sub-Agent 3 scores 6 signals, assigns tier, ranks → returns sorted scored list
7	Orchestrator takes top 10 (all tiers) → sends to Sub-Agent 4
8	Sub-Agent 4 builds top-3 trade cards (Tier A first, then B) with SL/TP and commentary
9	Orchestrator renders output to console — trade cards + rescan time if applicable
10	User reviews → instructs 'Enter trade [X] with £[N] risk' → Execution Agent fires

§2 — Sub-Agent 1: Data Retrieval Agent

Responsible for all MCP data calls. Operates sequentially per instrument to remain stable under the HTTP connection. Designed to be the most MCP-aware agent — it handles retries, empty responses, and symbol name mapping (Spread Betting instruments may use different naming conventions to CFD).

Data Fetched Per Instrument

Data	Depth	Used For
4H OHLCV	200 bars	Gate analysis, EMA stack, ADX, RSI divergence, Fibonacci range
1H OHLCV	100 bars	Entry timing, EMA proximity, RSI zone, ATR check, TF alignment
Live price (bid/ask)	Current tick	Trade card current price, SL/TP pip calculations
Symbol metadata	Once at scan start	Pip size, min volume, contract size for position sizing

MCP Connection

Use CTrader MCP via HTTP as primary. This is the proven stable route from existing agents.

```
Connection: cTrader MCP HTTP endpoint (same config as existing Claude Code agents)
Timeframe codes: H4 = '4H' or 'HOUR4' | H1 = '1H' or 'HOUR1' (confirm from MCP schema)
Bars request: getBars(symbol, timeframe, count) – request 210 bars, use last 200 to avoid incomplete current
```

Instrument Discovery

At scan start, the agent calls getSymbols (or equivalent MCP tool) to retrieve the full list of available Spread Betting instruments. It then merges this with the 33 pre-configured instruments, deduplicates, and filters to tradeable symbols only (excluding delisted or halted). The full merged list becomes the scan universe.

Account type flag: SPREAD_BETTING. If this is later changed to CFD, update the account type param — all other logic remains identical. This should be a config constant at the top of the agent.

Error Handling

Condition	Handling
MCP timeout on symbol	Retry once after 2s. If still fails, log 'DATA_FAIL: [symbol]' and skip
Empty bars returned	Log 'INSUFFICIENT_DATA: [symbol]' — skip. Do not attempt gate analysis on incomplete data
Symbol not found	Log 'SYMBOL_NOT_FOUND: [symbol]' — skip silently
Partial data (< 150 bars)	Skip — ADX/EMA calculations are unreliable on short series

§3 — Sub-Agent 2: Gate Layer Agent

Applies four binary gates to every instrument. All four must pass for an instrument to proceed to scoring. A single gate failure removes the instrument. Each failure is logged with the specific gate that failed so Pravinder can audit why a particular instrument was excluded.

The Four Gates

Gate	Name	TF	Condition	Rationale
G1	Trend Exists	4H	ADX(14) > 25 AND current ADX > previous ADX bar	Confirms trend in force and momentum building. Rising ADX = acceleration, not exhaustion.
G2	EMA Stack	4H	LONG: EMA21 > EMA50 > EMA200 SHORT: EMA21 < EMA50 < EMA200	Full alignment of short/mid/long MAs = institutional trend confirmed across all horizons.
G3	Price Side	4H	LONG: Close > EMA21 SHORT: Close < EMA21	Price must be on the correct side of the fast MA. Guards against counter-trend entries.
G4	No Divergence	4H	RSI(14) must NOT be diverging from price at recent swing highs/lows. Method: compare last 2 price peaks (troughs for short) against last 2 RSI peaks (troughs). If price HH + RSI LH = bearish divergence → FAIL for long. If price LL + RSI HL = bullish divergence → FAIL for short.	Divergence = momentum decay. This is the primary exhaustion guard.

Direction Assignment

Direction (LONG/SHORT) is determined by G2 (EMA stack direction) and confirmed by G3 (price side). These must agree. If EMA stack is bullish but price is below EMA21, G3 fails and the instrument is excluded. No ambiguous direction — each instrument exits this agent with a single, clear directional tag.

Gate Agent Output Schema

```
{
  "symbol": "UK100",
  "passed": true,
  "direction": "LONG",
  "gates": {
    "G1_adx": true,
    "G2_ema_stack": true,
    "G3_price_side": true,
    "G4_no_divergence": true
  },
  "adx_value": 31.4,
  "adx_rising": true
}
```

Swing Detection for G4 (Divergence Gate)

To find swing highs/lows for the divergence check, use a simple N-bar lookback:

- Swing high = bar where high > previous N bars and > following N bars. Use N=3.
- Identify the last 2 confirmed swing highs (or lows for short).
- Compare: if price_high_2 > price_high_1 (higher high) but rsi_high_2 < rsi_high_1 (lower RSI high) → divergence = TRUE → G4 FAIL.
- Only flag divergence if both swing highs are within last 50 bars. Older divergence is not actionable.

§4 — Sub-Agent 3: Scoring & Ranking Agent

Takes all gate-passed instruments and scores each across 6 signals. Signals are intentionally non-correlated — each captures a different dimension of trade quality. The total score determines tier assignment and ranking.

Scoring Matrix

#	Signal	TF	Max	Scoring	Rationale
S 1	ADX Strength	4H	20	ADX 25–29: 10pts ADX 30–39: 15pts ADX ≥ 40: 20pts	Higher ADX = stronger trend momentum. Most important signal.
S 2	1H EMA21 Pullback Proximity	1H	20	Price within 0.3% of 1H EMA21: 20pts Price within 0.75% of 1H EMA21: 10pts Beyond 0.75%: 0pts	Buying near EMA on the 1H = entry at value, not chase. Highest probability entry timing.
S 3	1H RSI Momentum Zone	1H	15	LONG: RSI(14) between 45–65: 15pts SHORT: RSI(14) between 35–55: 15pts Outside zone: 0pts	RSI in momentum zone = trend active, not overbought/oversold. Avoids extended entries.
S 4	ATR Range Guard	1H	15	Current 1H bar range < 1.0x ATR(14): 15pts Range < 1.5x ATR(14): 8pts Range ≥ 2x ATR(14): 0pts	Large bar range = potential exhaustion move. Entering after calm bars improves R:R.
S 5	Multi-Timeframe EMA Alignment	1H+4H	15	1H EMA stack ALSO aligned same direction as 4H: 15pts Not aligned: 0pts	Both timeframes in alignment = maximum trend consensus. Binary signal — full points or none.
S 6	Fibonacci Retracement Zone	4H	15	Price between 38.2%–61.8% retracement of last major swing leg: 15pts Between 23.6%–38.2% or 61.8%–76.4%: 8pts Outside these zones: 0pts Swing reference: highest high and lowest low of last 50 4H bars	Fib zone = statistically common institutional re-entry level. Adds structural confluence.

Tier Assignment

Tier	Score Range	Interpretation
A — Prime Setup	75–100	All or most conditions optimal. Highest conviction. Surface in output first.
B — Good Setup	50–74	Solid setup, minor conditions not ideal. Tradeable with standard risk.
C — Watch	30–49	Gates passed but entry timing weak. Logged but NOT shown in trade card output. Monitor for later rescan.

Ranking Output

The scoring agent returns a sorted list, Tier A first (highest score first within tier), then Tier B. Tier C instruments are included in a separate 'watchlist' array but not promoted to trade cards. The top 10 scored instruments (A+B combined) are passed to Sub-Agent 4. Sub-Agent 4 selects the top 3.

§5 — Sub-Agent 4: Trade Card & Commentary Agent

Receives the top-10 ranked instruments and produces the final output: three trade cards plus, where applicable, a 'watch' section for setups that are valid but not yet at entry price. This agent performs the SL and TP calculations, formats all output, and generates Claude's analytical commentary.

SL Calculation Logic

Stop loss is calculated using ATR-based methodology, anchored to structure:

- Base SL distance = $1.5 \times \text{ATR}(14)$ on the 4H timeframe.
- LONG: SL placed at the lower of: (entry price – $1.5 \times \text{ATR}4\text{H}$) OR (last 4H swing low – $0.5 \times \text{ATR}4\text{H}$ buffer).
- SHORT: SL placed at the higher of: (entry price + $1.5 \times \text{ATR}4\text{H}$) OR (last 4H swing high + $0.5 \times \text{ATR}4\text{H}$ buffer).
- The more conservative (wider) of the two values is always used — protects against being stopped on noise.

TP Calculation Logic

Three take profit levels using ATR multiples from entry (respects instrument volatility):

Level	Formula	Purpose
TP1	$1.5 \times \text{SL distance from entry}$	First partial — bank profit, reduce risk
TP2	$2.5 \times \text{SL distance from entry}$	Main target — favourable R:R
TP3	$4.0 \times \text{SL distance from entry}$	Runner target — let it trend

'Not At Price Yet' Logic

If the highest-scoring instruments are valid (gates passed, tier A or B) but price is NOT currently at the 1H EMA21 zone (S2 score = 0), the trade card is still generated but flagged as 'WATCH — ENTRY NOT YET REACHED'. The card includes:

- Current price vs entry zone (EMA21 level).
- How far price is from the EMA21 entry zone (in points and %).
- Recommended rescan time: next full 1H candle close after estimated pullback window (see §10).

Trade Card Format

Each card is output in the following structured format (rendered as formatted console output):

[illegible]

Watch Card Format (Not At Entry Yet)

[illegible][illegible]

§6 — Execution Agent: Order Placement

The Execution Agent is a separate, explicitly invoked agent. It only fires when Pravinder gives a direct instruction such as 'Enter trade 1 — UK100 with £450 risk'. It never executes autonomously. It confirms the trade parameters before placing, then executes via CTrader MCP.

Risk-Based Position Sizing (Spread Betting)

Spread betting uses stake per point (£/point) rather than lots. The Execution Agent calculates stake as:

SL distance (points) = Entry price – Stop Loss price (absolute value)
 Stake per point (£/pt) = Risk Amount (£) ÷ SL distance (points)
 Example: £450 risk, SL = 55 points → stake = £450 ÷ 55 = £8.18/pt
 Round down to nearest £0.50/pt increment for clean sizing

Pre-Execution Confirmation

Before placing the order, the agent outputs a confirmation block for Pravinder to verify:

```
■■ ORDER CONFIRMATION ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ Instrument : UK100 Direction : BUY
Entry : MARKET (current ask ~8,342) Stop Loss : 8,287 (55 pts) TP1 : 8,422 | TP2: 8,475 | TP3:
8,562 Risk : £450 Stake : £8.00/pt (rounded down from £8.18) Max loss : £440 (at SL) Type CONFIRM
to place, or CANCEL to abort. ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
```

Order Placement via MCP

On CONFIRM, the agent places a market order with SL and TP1 attached. TP2 and TP3 are placed as separate limit orders tied to the same position. This mirrors the cTrader native approach for multiple TPs.

Step	Action
1	Call MCP: createMarketOrder(symbol, direction, volume_in_stake, stopLoss, takeProfit=TP1)
2	Capture positionId from MCP response
3	Call MCP: createLimitOrder(symbol, CLOSE_direction, volume/3, TP2, linkedPositionId)
4	Call MCP: createLimitOrder(symbol, CLOSE_direction, volume/3, TP3, linkedPositionId)
5	Output: confirmed order details, position ID, entry price filled

Execution Error Handling

Condition	Handling
MCP order rejected	Report error message verbatim. Do not retry automatically. Ask Pravinder to review.
Stake below broker minimum	Alert: 'Stake EX.XX is below minimum — adjust risk or skip this trade'
Price moved beyond SL before fill	Alert: 'Price has moved through SL level — trade no longer valid. Abort.'
Partial fill	Report fill details. Ask whether to place remainder or cancel.

§7 — Gate Logic: Full Specification

Complete pseudocode for each gate. Claude Code should implement exactly as specified.

G1 — ADX Gate

```
# Inputs: 4H OHLCV bars (200 bars)
# Calculate ADX(14) across all bars
adx_current = adx[-1] # latest completed bar
adx_previous = adx[-2]
G1_pass = (adx_current > 25) AND (adx_current > adx_previous)
# Note: use standard Wilder smoothing for ADX calculation
# directional_movement: +DM and -DM from high/low differences
```

G2 — EMA Stack Gate

```
ema21 = EMA(close_4H, 21)[-1]
ema50 = EMA(close_4H, 50)[-1]
ema200 = EMA(close_4H, 200)[-1]
bull_stack = (ema21 > ema50) AND (ema50 > ema200)
bear_stack = (ema21 < ema50) AND (ema50 < ema200)
G2_pass = bull_stack OR bear_stack
direction = 'LONG' if bull_stack else 'SHORT'
```

G3 — Price Side Gate

```
close_4H_latest = bars_4H[-1].close
if direction == 'LONG':
    G3_pass = (close_4H_latest > ema21)
else:
    G3_pass = (close_4H_latest < ema21)
```

G4 — Divergence Gate

```
# Step 1: Calculate RSI(14) on 4H bars
rsi_4H = RSI(close_4H, 14)

# Step 2: Find last 2 swing highs (for LONG check)
# Swing high: bar[i].high > max(bar[i-3:i].high) AND bar[i].high > max(bar[i+1:i+4].high)
swing_highs = find_swings(bars_4H, type='high', n=3, lookback=50)[-2:]
swing_lows = find_swings(bars_4H, type='low', n=3, lookback=50)[-2:]

# Step 3: Check divergence
if direction == 'LONG' and len(swing_highs) >= 2:
    price_hh = swing_highs[-1].high > swing_highs[-2].high # higher high
    rsi_lh = rsi_4H[swing_highs[-1].idx] < rsi_4H[swing_highs[-2].idx] # lower RSI
    bearish_div = price_hh AND rsi_lh
    G4_pass = NOT bearish_div
elif direction == 'SHORT' and len(swing_lows) >= 2:
    price_ll = swing_lows[-1].low < swing_lows[-2].low # lower low
    rsi_hl = rsi_4H[swing_lows[-1].idx] > rsi_4H[swing_lows[-2].idx] # higher RSI
    bullish_div = price_ll AND rsi_hl
    G4_pass = NOT bullish_div
else:
    G4_pass = True # insufficient swings to check — pass by default
```

§8 — Scoring Logic: Full Specification

S1 — ADX Strength (max 20pts)

```
adx = adx_current # from Gate Layer payload
if adx >= 40: s1 = 20
elif adx >= 30: s1 = 15
elif adx >= 25: s1 = 10
else: s1 = 0 # shouldn't reach here (G1 gate)
```

S2 — 1H EMA21 Proximity (max 20pts)

```
ema21_1H = EMA(close_1H, 21)[-1]
current_price = latest_bid # live price from MCP
distance_pct = abs(current_price - ema21_1H) / ema21_1H * 100
if distance_pct <= 0.3: s2 = 20
elif distance_pct <= 0.75: s2 = 10
else: s2 = 0
```

S3 — 1H RSI Zone (max 15pts)

```
rsi_1H = RSI(close_1H, 14)[-1]
if direction == 'LONG':
    s3 = 15 if 45 <= rsi_1H <= 65 else 0
else: # SHORT
    s3 = 15 if 35 <= rsi_1H <= 55 else 0
```

S4 — ATR Range Guard (max 15pts)

```
atr_1H = ATR(bars_1H, 14)[-1]
current_bar_range = bars_1H[-1].high - bars_1H[-1].low
ratio = current_bar_range / atr_1H
if ratio < 1.0: s4 = 15
elif ratio < 1.5: s4 = 8
else: s4 = 0
```

S5 — Multi-TF Aligment (max 15pts)

```
ema21_1H = EMA(close_1H, 21)[-1]
ema50_1H = EMA(close_1H, 50)[-1]
ema200_1H = EMA(close_1H, 200)[-1]
bull_1H = (ema21_1H > ema50_1H) AND (ema50_1H > ema200_1H)
bear_1H = (ema21_1H < ema50_1H) AND (ema50_1H < ema200_1H)
if direction == 'LONG':
    s5 = 15 if bull_1H else 0
else:
    s5 = 15 if bear_1H else 0
```

S6 — Fibonacci Zone (max 15pts)

```
# Reference range: highest high and lowest low of last 50 4H bars
swing_high_ref = max(bar.high for bar in bars_4H[-50:])
swing_low_ref = min(bar.low for bar in bars_4H[-50:])
swing_range = swing_high_ref - swing_low_ref
fib_236 = swing_high_ref - 0.236 * swing_range
fib_382 = swing_high_ref - 0.382 * swing_range
fib_618 = swing_high_ref - 0.618 * swing_range
```

```
fib_764 = swing_high_ref - 0.764 * swing_range
if fib_382 <= current_price <= fib_618: s6 = 15
elif fib_236 <= current_price <= fib_382 or fib_618 <= current_price <= fib_764:
s6 = 8
else: s6 = 0
# Note: invert logic for SHORT (price below fib levels)
```

Total Score & Tier

```
total_score = s1 + s2 + s3 + s4 + s5 + s6 # max 100
tier = 'A' if total_score >= 75 else 'B' if total_score >= 50 else 'C'
```

§9 — Trade Card: Format & Output Specification

The full output of the agent is three trade cards rendered to console. The exact format is specified below. Claude Code must replicate this layout faithfully. Use Unicode box-drawing characters for borders.

Output Structure

Component	Content
Header block	Scan summary: instruments scanned, gates passed, scored, timestamp (UK time)
Trade Card 1	Top-ranked setup (Tier A preferred)
Trade Card 2	Second-ranked setup
Trade Card 3	Third-ranked setup
Watch List	Any Tier C instruments that passed gates — listed as single-line summaries, no full card
Rescan Guidance	If any cards are WATCH status — next recommended scan time in UK time with DST applied
Entry Prompt	Single line: 'To enter a trade, say: Enter trade [1/2/3] with £[amount] risk'

Scan Header Format

```
/TREND-CONTINUATION-AGENT | Scan complete
Time : 14:32 BST (Wed 11 Jun 2025)
Universe : 47 instruments scanned
Gates : 12 passed all 4 gates
Scored : 12 | Tier A: 3 | Tier B: 7 | Tier C: 2
Output : Top 3 trade cards below
```

§10 — Rescan Logic: UK Time & DST Awareness

The agent must always display times in UK local time, correctly applying DST. BST (British Summer Time, UTC+1) applies from the last Sunday in March to the last Sunday in October. GMT (UTC+0) applies outside this window. The agent should determine which is active at runtime.

Rescan Time Calculation

When a setup is valid but price is not yet at entry ($S2 = 0$), the agent calculates the recommended rescan time as:

- Find the current 1H candle close time (next top of hour).
- Recommend rescan at: `current_1H_close + 1 hour` (i.e., 2 candles forward).
- If the next 1H close is within 10 minutes, recommend the following close instead.
- Always display in UK time with timezone label: 'BST' or 'GMT'.

Market Session Context

The agent should note the current session in its commentary where relevant:

Session	UK Time	Commentary Note
Pre-London	06:00–08:00 BST/GMT	Low liquidity. Flag if trade card fires in this window — wider spreads.
London Open	08:00–10:00 BST/GMT	High volatility, strong continuation moves. Best session for index trends.
London Mid	10:00–14:00 BST/GMT	Steadier trending. Strong setups here are high quality.
NY Overlap	14:00–17:00 BST/GMT	Maximum liquidity. Trend continuation most reliable.
NY Only	17:00–21:00 BST/GMT	US indices relevant. UK100/GER40 lower volume after 17:30.
After Hours	21:00–06:00 BST/GMT	Flag explicitly. Do not recommend entry during this window.

§11 — CTrader MCP HTTP: Connection & Instrument Discovery

All data and execution calls use the CTrader MCP via HTTP. This is the same connection pattern used in Pravinder's existing Claude Code agents and is the proven stable route.

Connection Configuration

```
# Connection type: HTTP (not WebSocket)
# Use same endpoint config as existing agents in this Claude Code session
# Account type filter: SPREAD_BETTING
# CFD switch: change account type param only – all logic identical
```

Key MCP Operations Used

MCP Tool	Purpose
getSymbols / listSymbols	Instrument discovery — fetch full tradeable symbol list
getBars / getHistoricalBars	OHLCV data — 4H (200 bars) and 1H (100 bars) per symbol
getTick / getLivePrice	Current bid/ask for trade card price display and SL/TP calc
getSymbolInfo	Pip size, min stake, contract size for position sizing
createMarketOrder	Execute trade at market with SL and TP1
createLimitOrder	Place TP2 and TP3 as separate limit close orders
getPositions	Confirm position opened successfully post-execution
getAccountInfo	Balance, equity, margin check before execution

Instrument Naming — Spread Betting

Pepperstone Spread Betting instrument names may differ slightly from CFD names. The agent should use the MCP symbol list as the canonical source of truth for names. The 33 pre-configured instruments should be mapped to their SB equivalents at startup using a name-matching function (fuzzy match on base name: 'UK100', 'GER40', etc.).

If a pre-configured symbol is not found in the SB instrument list, log a warning and skip it — do not fail the entire scan.

§12 — Implementation Notes for Claude Code

Critical Implementation Rules

- Use CTrader MCP via HTTP as the ONLY data and execution route. Do not attempt WebSocket.
- All indicator calculations (ADX, EMA, RSI, ATR) must be implemented from scratch using OHLCV arrays — do not rely on MCP indicator endpoints as these vary by platform version.
- ADX calculation: use Wilder's smoothing (RMA, not EMA). Standard EMA will produce different values.
- EMA calculation: standard exponential smoothing, multiplier = $2/(N+1)$. Seed with SMA for first N bars.
- RSI calculation: use Wilder's RSI (RMA of gains/losses). Standard RSI libraries use the same method.
- ATR calculation: true range = $\max(\text{high-low}, \text{abs}(\text{high-prev_close}), \text{abs}(\text{low-prev_close}))$. ATR = RMA of TR over N bars.
- Bar indexing: [-1] = latest COMPLETED bar. Do not use the current live bar for calculations — it is incomplete.
- Never execute an order without the explicit confirmation step. The agent must pause and wait for 'CONFIRM'.
- Always display times in UK local time with BST/GMT label. Never display UTC without conversion.
- Stake rounding: always round DOWN to protect against exceeding stated risk. Never round up.

File & Folder Structure

```
/trend-continuation-agent/  
orchestrator.py # Main entry point  
agents/  
data_retrieval.py # Sub-Agent 1  
gate_layer.py # Sub-Agent 2  
scoring_ranking.py # Sub-Agent 3  
trade_card.py # Sub-Agent 4  
execution.py # Execution Agent  
utils/  
indicators.py # ADX, EMA, RSI, ATR from scratch  
time_utils.py # UK time / DST conversion  
position_sizing.py # Stake calculation for SB  
config.py # Instrument list, account type, thresholds
```

Confia Constants (confia.py)

```
ACCOUNT_TYPE = 'SPREAD_BETTING' # Change to 'CFD' when needed  
MCP_CONNECTION = 'HTTP'  
BARS_4H = 210 # fetch 210, use last 200 (skip live incomplete bar)  
BARS_1H = 110 # fetch 110, use last 100  
ADX_PERIOD = 14  
EMA_FAST = 21 | EMA_MID = 50 | EMA_SLOW = 200  
RSI_PERIOD = 14  
ATR_PERIOD = 14  
SWING_N = 3 # bars each side for swing detection  
SWING_LOOKBACK = 50 # max bars back for divergence check  
GATE_ADX_MIN = 25  
TIER_A_MIN = 75 | TIER_B_MIN = 50  
SL_ATR_MULTIPLIER = 1.5  
TP1_R = 1.5 | TP2_R = 2.5 | TP3_R = 4.0
```

Pre-Configured Instrument List (33 instruments)

The following is the existing scan universe. The agent extends this with any additional SB instruments returned by the MCP symbol discovery call.

UK100	GER40	US500	US30	NAS100	US2000
AUS200	JPN225	FRA40	EU50	HK50	XAUUSD
XAGUSD	USOIL	UKOIL	EURUSD	GBPUSD	USDJPY
AUDUSD	USDCAD	USDCHF	NZDUSD	EURGBP	EURJPY
GBPJPY	AUDJPY	CADJPY	BTCUSD	ETHUSD	XPDUSD
XPTUSD	NATGAS	CORN			

Trigger Command

The agent is invoked in Claude Code with:

```
/Trend-Continuation-Agent
```

Optionally with a specific instrument override:

```
/Trend-Continuation-Agent --symbols UK100,GER40,US500
```

Rescan After Trade Entry

After executing a trade, the agent should note: 'Rescan available at any time with /Trend-Continuation-Agent — open positions will be visible in the execution summary.' The agent does not automatically monitor open positions — that remains Pravinder's discretion.